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Research Interest

- Multimodal Industrial Vision (Experienced)
- Medical Image Analysis (Experienced)
- Vision-Language-Action Models (Rookie)

Publications

[†] *Equal Contribution*, ^{*} *Corresponding Author*

- [1] **Shuai Jiang**, Yunfeng Ma, Jingyu Zhou, Yaonan Wang, and Min Liu*, [Unified Multimodal Industrial Anomaly Detection via Few Normal Samples](#), *IEEE Transactions on Industrial Informatics (T-II)*, 2026. (第一作者)
- [2] **Shuai Jiang**[†], Yunfeng Ma[†], Jingyu Zhou, Yuan Bian, Yaonan Wang, and Min Liu*, [Resilient Multimodal Industrial Surface Defect Detection with Uncertain Sensors Availability](#), *IEEE/ASME Transactions on Mechatronics (T-Mech)*, 2025. (共同第一作者)
- [3] Kaijie Li[†], **Shuai Jiang**[†], Yao Lin, Jingyu Zhou, Hao Qu, Xiangfei Meng, Yaonan Wang, and Min Liu*, [Micro Surface Defect Inspection of Aero-Engine Blades via Dynamic Cross-Scale Semantic Aggregation](#), *IEEE Transactions on Industrial Informatics (T-II)*, 2026. (共同第一作者)
- [4] **Shuai Jiang**, Kai Hu*, Yuan Zhang, and Xieping Gao, [Joint Segmentation of FAZ and RVs in OCTA Images with Auxiliary 3D Image Projection Learning](#), *IEEE Transactions on Instrumentation and Measurement (T-IM)*, 2024. (第一作者)

- [5] **Shuai Jiang**, Min Liu*, Yuxi Liu, Yunfeng Ma, and Yaonan Wang, [Category-agnostic Cluster-guided Selective Suppression for Weakly Supervised Surface Defect Localization](#), *IEEE Transactions on Instrumentation and Measurement (T-IM)*, 2024. (第一作者)
- [6] Kai Hu, **Shuai Jiang**, Dong Liu, and Xieping Gao*, [Segmentation of Retinal Layer Boundary in OCT Images Based on End-to-end Deep Neural Network and Graph Search](#), *Journal of Software (软件学报)*, 2023. (学生第一作者)
- [7] Kai Hu, **Shuai Jiang**, Yuan Zhang, Xuanya Li, and Xieping Gao*, [Joint-Seg: Treat Foveal Avascular Zone and Retinal Vessel Segmentation in OCTA Images as a Joint Task](#), *IEEE Transactions on Instrumentation and Measurement (T-IM)*, 2022. (学生第一作者)
- [8] Yunfeng Ma, Min Liu*, **Shuai Jiang**, Jingyu Zhou, Yuan Bian, Xueping Wang, and Yaonan Wang, [ZUMA: Training-free Zero-shot Unified Multimodal Anomaly Detection](#), *IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)*, 2026. (其他作者)
- [9] Jingyu Zhou, Yunfeng Ma, **Shuai Jiang**, Yaonan Wang, and Min Liu*, [Cross-View Dynamic Learning-Based Multi-Class Industrial Anomaly Detection](#), *IEEE Transactions on Circuits and Systems for Video Technology (T-CSVT)*, 2026. (其他作者)
- [10] Jingyu Zhou, Yunfeng Ma, **Shuai Jiang**, Yaonan Wang, and Min Liu*, [SCAP: Semantic Prototype Alignment for Robust Point Cloud Registration](#), *IEEE Transactions on Circuits and Systems for Video Technology (T-CSVT)*, 2026. (其他作者)
- [11] Meng Gao†, Yunfeng Ma†, **Shuai Jiang**, Qi Su, Xiangfei Meng, Yaonan Wang, and Min Liu*, [PANDA: Progressive Adaptive Network for Defect-aware Few-shot Segmentation](#), *IEEE Transactions on Industrial Informatics (T-II)*, 2025. (其他作者)
- [12] Yunfeng Ma, Min Liu*, **Shuai Jiang**, Jingyu Zhou, Yuan Bian, and Yaonan Wang, [Efficient Feature Coupling for Industrial Few-shot Defect Segmentation](#), *IEEE/ASME Transactions on Mechatronics (T-Mech)*, 2025. (其他作者)
- [13] Yuxi Liu†, Min Liu†, **Shuai Jiang**, Yi Tang*, and Yaonan Wang, [Searching Efficient Semantic Segmentation Architectures via Dynamic Path Selection](#), *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. (其他作者)
- [14] Yunfeng Ma, Min Liu*, **Shuai Jiang**, Xueping Wang, Yuan Bian, and Yaonan Wang, [Multi-Context Aggregation Network with Foreground Correction for Automated Few-Shot Defect Segmentation](#), *IEEE Transactions on Automation Science and Engineering (T-ASE)*, 2025. (其他作者)
- [15] Zekai Yao†, Jiazheng Wang†, Yunfeng Ma, **Shuai Jiang**, Yingmei He, Yaonan Wang, and Min

- Liu*, [Edge-guided Semantic Cross-Matching Network for Few-Shot Strip Steel Surface Defect Segmentation](#), *IEEE Transactions on Instrumentation and Measurement (T-IM)*, 2026. (其他作者)
- [16] Jingyu Zhou, Min Liu*, Yunfeng Ma, **Shuai Jiang**, and Yaonan Wang, [Multi-View Attention Guided Feature Learning for Unsupervised Surface Defect Detection](#), *IEEE/ASME Transactions on Mechatronics (T-Mech)*, 2025. (其他作者)
- [17] Yuan Bian, Min Liu*, Yunqi Yi, Xueping Wang, **Shuai Jiang**, Yaonan Wang, [Prompt-driven Transferable Adversarial Attack on Person Re-Identification with Attribute-aware Textual Inversion](#), *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. (其他作者)
- [18] Yuxi Liu†, Yunfeng Ma†, Yi Tang, Min Liu*, **Shuai Jiang**, and Yaonan Wang, [Automated Neural Architecture Design for Industrial Defect Detection](#), *IEEE/ASME Transactions on Mechatronics (T-Mech)*, 2025. (其他作者)
- [19] Yaoyi Cai, Dan Huang, ZhiXun Li, Yunfeng Ma*, **Shuai Jiang**, Zekai Yao*, and Min Liu*, [EGCR-Net: Edge-Guided Cross-Region Network for Semi-Supervised Soldering Defect Detection](#), *IEEE Transactions on Instrumentation and Measurement (T-IM)*, 2025. (其他作者)
- [20] Jingyu Zhou†, Yunfeng Ma†, Zhenhai Li, Min Liu*, **Shuai Jiang**, and Yaonan Wang, [3D Imaging Optical Path-Guided Viewpoint Planning Method for Free-Form Surface Inspection](#), *IEEE/ASME Transactions on Mechatronics (T-Mech)*, 2025. (其他作者)
- [21] Xihang Cheng†, Xiangfei Meng†, Min Liu*, Yuxi Liu, Yunfeng Ma, **Shuai Jiang**, and Yaonan Wang, [Neural Architecture Search-based Detection Method and System for Aero-engine Blade Surface Defects](#), *SCIENTIA SINICA Informationis (中国科学：信息科学)*, 2025. (其他作者)

Education

[†] Indicates expected

Sept. 2023 – Dec. 2026 [†] Ph.D., Control Science and Engineering, *Hunan University*

Supervisors: [Min Liu](#) and [Yaonan Wang](#)

Oct. 2020 – Jun. 2023 M.Eng., Computer Technology, *Xiangtan University*

Supervisors: [Kai Hu](#) and [Xieping Gao](#)

Sept. 2016 – Jun. 2020 B.Sc., Network Engineering, *Xiangtan University*

Selected Honours and Awards

2025 GENCA Scholarship, **Top 1%**, *Hunan University*

2023 Excellent Graduates, **Top 1%**, *Hunan Provincial Department of Education*

2023 Principal Scholarship, **Top 1%**, *Xiangtan University*

2022 China National Scholarship, **Top 1.2%**, *Ministry of Education of the People's Republic of China*

2022 Special Scholarship, **Top 5%**, *Xiangtan University*

2021 Special Scholarship, **Top 5%**, *Xiangtan University*

2020 Third Prize of ISICDM Challenges, **Rank #7**, *ISICDM 2020*

Academic Services

Journal Reviewer

- IEEE Transactions on Industrial Informatics, Automation in Construction, Scientific Reports, IEEE Access, Robot Learning

Conference Reviewer

- NeurIPS, CVPR, ICCV, AAAI, ACMMM, ICASSP, ICME